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Sparse Recovery and Compressed Sensing: A Model-Based Deep Learning Approach

From ISTA to HyperLISTA — what is worth learning in a deep-unrolled sparse-recovery network?

Final Project Report

Model-Based Deep Learning (361.2.2320) — Spring 2026

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Abstract

We study deep unrolling of ISTA for sparse recovery across the full spectrum of learned flexibility — from classical ISTA/FISTA (no learning), through our three original ISTA-derived scalar models (16–32 parameters) and LISTA ($\sim$6M parameters), to ALISTA (32) and HyperLISTA (3 tuned scalars). On synthetic Gaussian-sparse data the model-based hierarchy is striking: HyperLISTA attains -61.4 dB NMSE with three grid-searched scalars, and our StepThresholdISTA outperforms full LISTA with 187,000$\times$ fewer parameters — the step-size schedule is the single most valuable component to learn. We also compare training strategies for unfolded optimizers (end-to-end, deep supervision, greedy layer-wise, weight tying). On pixel-domain compressed sensing of MNIST and Fashion-MNIST the ranking inverts: methods built on the Gaussian-sparse prior collapse while data-driven LISTA remains robust — structure wins when the model matches the data, and a wrong prior is worse than no prior. Repeating the image experiments in the DCT domain confirms the diagnosis both ways: structured methods recover where pixels are not sparse (HyperLISTA on Fashion-MNIST: $-0.3 \to -7.7$ dB) and degrade where they already are (MNIST at ratio 0.5) — the recovery domain is itself part of the signal model.

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1 Motivation and Background

Many acquisition systems observe a signal only through a small number of linear measurements: accelerated MRI samples a fraction of k-space, radar receives compressed snapshots, and single-pixel cameras record random projections. Recovering the underlying signal from such incomplete data is an ill-posed linear inverse problem — infinitely many signals explain the same measurements. *Compressed sensing* resolves this ambiguity with a structural prior: natural signals are (approximately) *sparse* in a suitable domain, i.e., only $s \ll n$ of their coefficients are significant [1].

Classical sparse-recovery algorithms such as ISTA and FISTA [2] solve a convex relaxation of the problem with provable convergence, and they are fully interpretable — but they may require hundreds of iterations to reach useful accuracy. Deep neural networks, at the other extreme, produce excellent reconstructions in a fixed (small) number of layers, but discard the known measurement model and require large amounts of data.

Model-based deep learning (MBDL) [7] bridges the two: the iterative algorithm is *unrolled* into a fixed-depth network whose layers retain the algorithmic structure, and only selected components are learned from data. The seminal example is LISTA [3], which unrolls ISTA and learns its matrices per layer. A later line of work progressively *re-introduced structure*: LISTA with coupled weights and support selection [4], ALISTA with analytically computed weights [5], and finally HyperLISTA [6], which reduces the entire network to *three* tuned scalars.

This project implements and evaluates this full spectrum. Our own contributions are: **(i)** a controlled *design-space study* built on three original ISTA-derived scalar models — ThresholdISTA, StepISTA and StepThresholdISTA — that isolate *which components of ISTA are actually worth learning*; **(ii)** a systematic comparison of the training strategies for unfolded optimizers discussed in class (end-to-end, deep supervision, greedy layer-wise, weight tying); and **(iii)** a cross-dataset robustness study of all methods on real image data under compressed sensing.

2 Problem Formulation

Measurement model. We observe

$$b = Ax^* + \varepsilon, \qquad A \in \mathbb{R}^{m\times n},\ m < n, \tag{1}$$

where $x^* \in \mathbb{R}^n$ is s-sparse ($\|x^*\|_0 \le s$), A is a known sensing matrix with unit-norm columns, and $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ is additive noise. The goal is to estimate x^* from (b, A).

Convex relaxation. The standard estimator is the LASSO (Least Absolute Shrinkage and Selection Operator),

$$\hat{x} \ = \ \arg\min_x \ \tfrac{1}{2}\|b - Ax\|_2^2 \ + \ \lambda\|x\|_1\,, \tag{2}$$

whose proximal-gradient iteration is ISTA:

$$x^{(k+1)} \ = \ \mathcal{S}_{\lambda/L}\Big(x^{(k)} + \tfrac{1}{L}A^T\big(b - Ax^{(k)}\big)\Big), \qquad L = \|A\|_2^2\,, \tag{3}$$

where $\mathcal{S}_\theta(v) = \mathrm{sign}(v)\max(|v| - \theta, 0)$ is the soft-thresholding operator. FISTA adds Nesterov momentum, improving the convergence rate from $O(1/k)$ to $O(1/k^2)$ [2].

Unrolled network. Fixing a budget of K iterations, an unrolled network is a map $x^{(K)} = f_\Theta(b)$ obtained by executing K steps of the form (3) in which selected quantities (step sizes, thresholds, matrices) become the trainable parameters Θ. Training minimizes the reconstruction error over a dataset of pairs $\{(b_i, x_i^*)\}$ generated from (1). Throughout this work $K = 16$.

Tasks and metrics. *Part A (synthetic):* $m = 250$, $n = 500$, $s = 50$, Gaussian A, noiseless measurements; 51,200 training, 2,048 validation and 2,048 test signals with $\mathcal{N}(0, 1)$ non-zeros on random supports. Performance is measured by normalized MSE in dB, $\mathrm{NMSE} = 10\log_{10}\mathbb{E}\big[\|\hat{x} - x^*\|_2^2 \,/\, \|x^*\|_2^2\big]$. *Part B (images):* pixel-domain compressed sensing of the MNIST and Fashion-MNIST datasets (28×28 grayscale images; 60,000 training / 10,000 test images each, of which 5,000 training images are held out for validation): the image is flattened to $x \in [0, 1]^{784}$ and measured as $b = Ax$ with measurement ratio $m/784 \in \{0.25, 0.5\}$. Both datasets are approximately sparse in the pixel domain (MNIST $\approx$ 81%, Fashion-MNIST $\approx$ 50% exact zeros), so we first attempt recovery directly in pixel space. Section 4.3 then repeats the experiment in the DCT domain: with the *same* measurements $b = Ax$, each model receives the effective dictionary $D = A\Psi^T$ (Ψ the orthonormal 2D-DCT basis), recovers the coefficients $\alpha = \Psi x$, and the image is $\hat{x} = \Psi^T\hat{\alpha}$. Orthonormality gives $\|D\|_2 = \|A\|_2$ (step sizes remain valid, no column renormalization) and NMSE on α equal to pixel NMSE, so the two domains are directly comparable. Metrics: NMSE, PSNR and SSIM. Benchmarks in both parts: ISTA and FISTA (classical), LISTA (fully learned), ALISTA and HyperLISTA (structured).

3 Method

3.1 The LISTA family

LISTA [3] unrolls (3) and learns all update matrices per layer:

$$x^{(k+1)} = \mathcal{S}_{\theta_k}\big(W_x^{(k)}x^{(k)} + W_y^{(k)}b\big), \qquad \Theta = \{W_x^{(k)} \in \mathbb{R}^{n\times n},\ W_y^{(k)} \in \mathbb{R}^{n\times m},\ \theta_k\}_{k=1}^K, \tag{4}$$

totaling $K(n^2+nm+1) \approx 6 \times 10^6$ parameters in our setting. Each layer is initialized at the ISTA parameterization ($W_y = \frac{1}{L}A^T$, $W_x = I - \frac{1}{L}A^TA$, $\theta = \lambda/L$), so the untrained network reproduces ISTA exactly. We also evaluate a *tied* variant sharing a single (W_x, W_y, θ) across layers (375K parameters).

ALISTA [5] shows the matrices need not be learned at all: a weight matrix W is computed analytically from A (we use the minimum-Frobenius solution $W = (AA^T)^{-1}A$, column-normalized so $\mathrm{diag}(W^TA) = 1$), and only per-layer scalars remain:

$$x^{(k+1)} = \mathcal{S}^{p_k}_{\theta_k}\big(x^{(k)} + \gamma_k W^T(b - Ax^{(k)})\big), \qquad \Theta = \{\gamma_k, \theta_k\}_{k=1}^K \;\; (2K = 32), \tag{5}$$

where $\mathcal{S}^p_\theta$ is soft-thresholding with *support selection*: the p largest-magnitude entries bypass shrinkage, with p ramping up across layers.

HyperLISTA [6] eliminates even the per-layer scalars. With μ the mutual coherence of W^TA and A^+ the pseudo-inverse, each layer computes its parameters *adaptively from the current residual*, driven by three global constants (c_1, c_2, c_3):

$$\theta^{(k)} = c_1\mu \left\|A^+(Ax^{(k)} - b)\right\|_1, \quad \beta^{(k)} = c_2\mu \left\|x^{(k)}\right\|_0, \quad p^{(k)} = c_3 \log \frac{\|A^+b\|_1}{\left\|A^+(Ax^{(k)} - b)\right\|_1}, \tag{6}$$

with a heavy-ball momentum term $\beta^{(k)}(x^{(k)} - x^{(k-1)})$ added to the update. Because only three scalars remain, they are found by a two-stage (coarse-to-fine) grid search on a validation set — *no backpropagation at all.* In our implementation the per-sample $\theta^{(k)}$ and $p^{(k)}$ are aggregated by the batch median, a simplification of the per-sample rule of [6].

3.2 Our design-space study: ISTA-derived scalar models

Between the extremes of LISTA and HyperLISTA we ask: *what is the minimal useful set of learnable components?* Generalizing (3) to $x^{(k+1)} = \mathcal{S}_{\theta_k}(x^{(k)} + \gamma_k A^T(b - Ax^{(k)}))$ exposes two scalar sequences, and we learn each subset in isolation:

Model	Learns	Coupling	#Params
ThresholdISTA	$\theta_1, \ldots, \theta_K$	step fixed at $1/L$	$K = 16$
StepISTA	$\gamma_1, \ldots, \gamma_K$	threshold tied, $\theta_k = \lambda\gamma_k$	$K = 16$
StepThresholdISTA	γ_k and θ_k	independent	$2K = 32$

All parameters are softplus-reparameterized to remain positive and initialized at the exact ISTA values, so every model starts from the ISTA baseline and training can only reallocate, never destroy, the algorithmic prior.

3.3 Training strategies for unfolded optimizers

Following the strategies discussed in class for training unfolded optimizers [8], we compare, for a fixed LISTA architecture:

- **L1 — end-to-end:** $\mathcal{L} = \left\|x^{(K)} - x^*\right\|^2$, standard BPTT through all K layers;
- **L2 — deep supervision:** $\mathcal{L} = \left\|x^{(K)} - x^*\right\|^2 + \lambda_{\text{int}} \sum_k w_k \left\|x^{(k)} - x^*\right\|^2$ with log-scaled layer weights w_k;

- **L3 — greedy layer-wise:** layer k is trained to minimize $\left\|x^{(k)} - x^*\right\|^2$ while layers $1, \ldots, k-1$ are frozen;

and additionally **weight tying** (shared layer parameters, RNN-style). Learned models are trained with Adam, early stopping on validation NMSE, and gradient clipping; HyperLISTA is tuned by grid search only.

4 Results

4.1 Part A: synthetic sparse recovery

Table 1 and Figure 1 summarize the main comparison at $K = 16$ layers. The model-based hierarchy is unambiguous: every increase in structure buys accuracy *and* removes parameters. HyperLISTA reaches −61.4 dB — near-perfect recovery — with three scalars and no backpropagation.

Table 1: Part A — synthetic sparse recovery (m=250, n=500, s=50, noiseless, K=16).

Method	NMSE [dB]	#Params	Comment
ISTA	−5.3	0	classical baseline
FISTA	−11.0	0	+ Nesterov momentum
ThresholdISTA (ours)	−7.1	**16**	thresholds alone barely help
StepISTA (ours)	−20.0	**16**	matches full LISTA
StepThresholdISTA (ours)	**−23.6**	**32**	beats LISTA, 187,000× fewer params
LISTA-L1	−20.3	6,000,016	fully learned matrices
LISTA-L2	−21.9	6,000,016	best LISTA variant
LISTA-L3	−16.6	6,000,016	greedy plateaus
LISTA-Tied-L1	−19.1	375,001	16× fewer params than LISTA
ALISTA	−29.8	32	analytic W
HyperLISTA	**−61.4**	**3**	grid search only

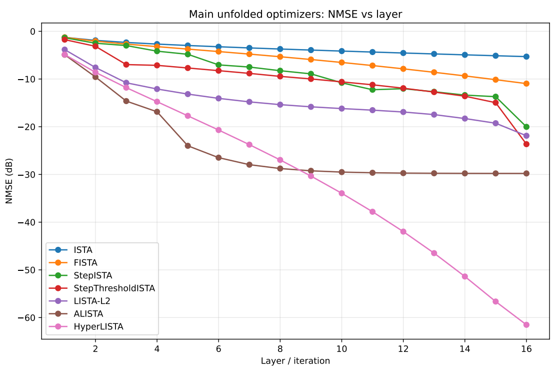

Figure 1: NMSE versus unfolded layer for the main models (Part A). HyperLISTA converges essentially linearly in depth; ALISTA saturates at −30 dB; learned-matrix models are slower per layer despite vastly more parameters.

Which ISTA components are worth learning? Figure 2 isolates the contribution of each scalar family. Learning thresholds alone (ThresholdISTA) yields only +1.8 dB over ISTA: with the conservative step $1/L$, the iterates change too slowly for threshold tuning to matter. Learning the *step-size schedule* alone (StepISTA, 16 scalars) is transformative — -20 dB, on par with 6M-parameter LISTA — because a learned, aggressive schedule (large early steps, decaying later) compensates for the finite depth. Decoupling the threshold from the step (StepThresholdISTA) adds another 3.6 dB at negligible cost. **The step-size schedule is the bottleneck component; matrix learning is largely redundant when A is fixed.**

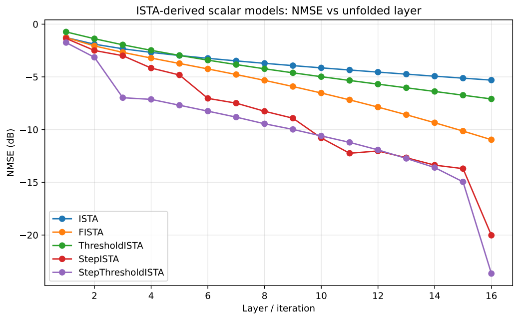

Figure 2: ISTA-derived scalar models (Part A). StepISTA and StepThresholdISTA accelerate sharply in the final layers — the learned schedule exploits the fixed depth budget.

Training strategies. Deep supervision (L2) is the best training method for untied LISTA (-21.9 vs. -20.3 dB for L1): intermediate losses stabilize gradients through 16 layers. Greedy layer-wise training (L3) underperforms (-16.6 dB), flattening from roughly layer 8 (Figure 3, left) — each greedy stage optimizes its own output, leaving later layers no incentive to improve. Weight tying loses only 1.2 dB relative to untied LISTA at 16× fewer parameters, acting as implicit regularization. Figure 3 (right) maps the design space by accuracy versus parameters: our scalar models and HyperLISTA occupy the efficient frontier, while all 6M-parameter LISTA variants cluster at the far right with no accuracy advantage.

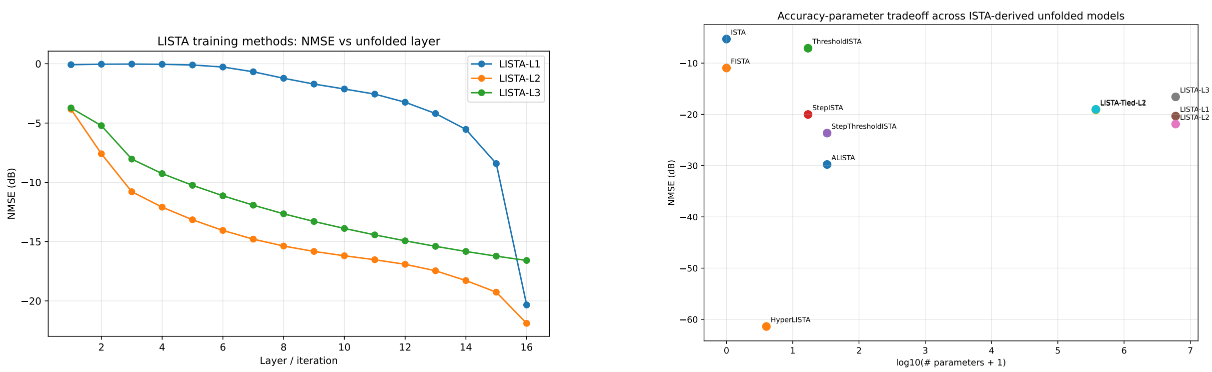

Figure 3: Left: LISTA training strategies (Part A). End-to-end L1 improves mostly in the final layers, deep supervision L2 improves every layer and ends best, and greedy L3 plateaus early. **Right:** accuracy–parameter trade-off across all evaluated models (bottom-left is ideal). Our StepISTA/StepThresholdISTA and HyperLISTA define the efficient frontier.

4.2 Part B: pixel-domain compressed sensing of real images

Table 2 reports the Fashion-MNIST results at measurement ratio 0.25, and Figure 5 shows reconstructions. The ranking *inverts*: the structured methods that dominated Part A now fail — ALISTA and HyperLISTA fall below plain ISTA — while LISTA-L2 reconstructs cleanly (SSIM 0.83).

Table 2: Part B — Fashion-MNIST pixel-domain CS (ratio 0.25, K=16).

Method	NMSE [dB]	PSNR [dB]	SSIM	#Params
ISTA	−1.1	8.8	0.13	0
FISTA	−1.1	8.7	0.13	0
StepThresholdISTA (ours)	−1.1	8.8	0.14	32
ALISTA	−0.8	8.4	0.13	32
HyperLISTA	−0.3	7.7	0.09	3
LISTA-L2	**−14.1**	**22.5**	**0.83**	12M

The cause is a *signal-model mismatch*. ALISTA's and HyperLISTA's guarantees are derived for exactly-sparse signals (at most s non-zeros with bounded magnitudes), and their adaptive rules were calibrated on i.i.d. Gaussian s-sparse data. Image pixels are bounded in $[0, 1]$, non-negative, spatially correlated, and only "soft-sparse." HyperLISTA's residual-driven threshold $\theta^{(k)} = c_1\mu \left\| A^{+}(Ax^{(k)} - b) \right\|_1$, calibrated for sparse Gaussian residuals, instead zeroes out valid image content. LISTA wins precisely because it makes no such assumption — with 55K training images it learns the actual pixel distribution.

Sweeping both datasets and both measurement ratios confirms the pattern (Figure 4): on MNIST at ratio 0.5 (an easy, genuinely sparse task) all methods are reasonable and the structured models degrade gracefully; as the data grows more textured (Fashion-MNIST) and measurements fewer (ratio 0.25), every prior-based method collapses while LISTA-L2 maintains SSIM ≥ 0.82 across all four scenarios.

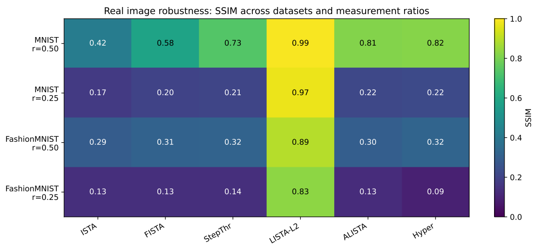

Figure 4: Robustness sweep (Part B): SSIM across {MNIST, Fashion-MNIST} × ratio {0.25, 0.5}. LISTA-L2 stays high in all four scenarios; the Gaussian-sparse-prior methods degrade sharply with data complexity and measurement scarcity.

4.3 Part B revisited: transform-domain recovery

We then test directly whether the collapse above is a *domain* problem or an *algorithm* problem: keeping the exact same measurements $b = Ax$ (same seed, sensing matrix and data splits as Table 2), all six methods are re-run over the effective dictionary $D = A\Psi^T$ of Section 2. Table 3 shows Fashion-MNIST at ratio 0.25 — the cell that broke every structured model.

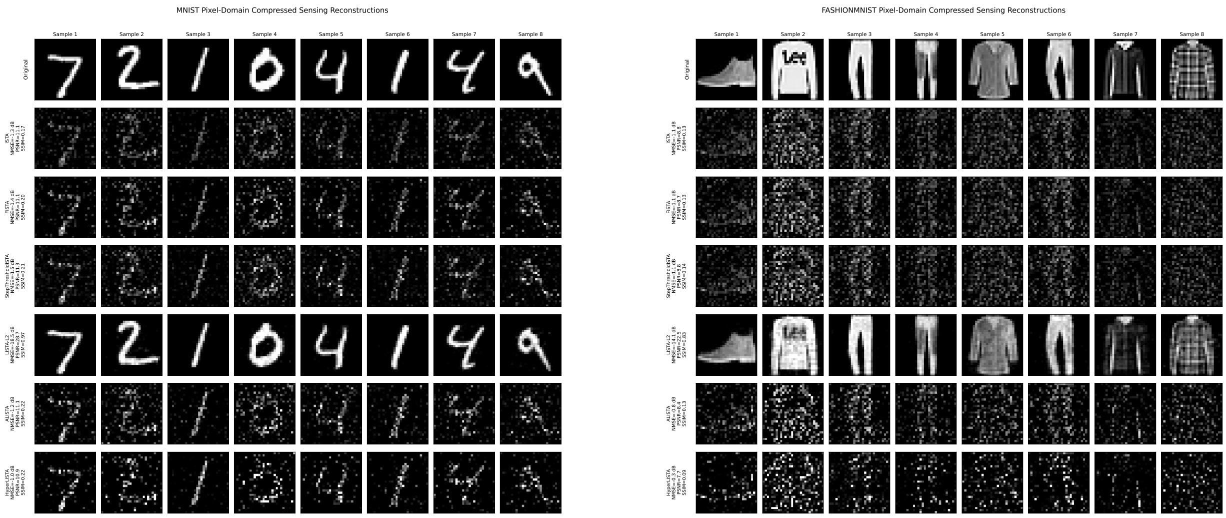

Figure 5: Visual stress test at ratio 0.25 (left: MNIST, right: Fashion-MNIST; rows: ground truth, ISTA, FISTA, StepThresholdISTA, LISTA-L2, ALISTA, HyperLISTA). LISTA-L2 recovers clean structure on both datasets; the Gaussian-sparse-calibrated thresholds of ALISTA/HyperLISTA erase valid signal on Fashion-MNIST.

Table 3: Part B revisited — same measurements as Table 2, recovery in the DCT domain (Fashion-MNIST, ratio 0.25, K=16). Arrows compare pixel-domain → DCT-domain values.

Method	NMSE [dB]	SSIM	#Params
ISTA	$-1.1 \rightarrow -4.0$	$0.13 \rightarrow 0.26$	0
FISTA	$-1.1 \rightarrow -5.6$	$0.13 \rightarrow 0.35$	0
StepThresholdISTA (ours)	$-1.1 \rightarrow -7.7$	$0.14 \rightarrow 0.52$	32
ALISTA	$-0.8 \rightarrow -5.1$	$0.13 \rightarrow 0.40$	32
HyperLISTA	$-0.3 \rightarrow -7.7$	$0.09 \rightarrow 0.52$	3
LISTA-L2	$\mathbf{-14.1 \rightarrow -14.9}$	$\mathbf{0.83 \rightarrow 0.81}$	12M

Every structured method recovers substantially (Figure 6). HyperLISTA climbs from dead last (−0.3 dB, SSIM 0.09) to −7.7 dB / SSIM 0.52 with the same three scalars, overtaking classical ISTA/FISTA and matching StepThresholdISTA; its tuned c_3 returns to a healthy support-selection range (2.5–15.8 across cells, versus 0.17 in pixel space) and grows with the measurement ratio, as the theory intends. At ratio 0.5 on Fashion-MNIST, HyperLISTA becomes the *best* structured method (−11.4 dB, SSIM 0.71). At ratio 0.25 (Table 3), LISTA-L2's NMSE lead over the best structured method shrinks from 13.0 dB (pixel) to 7.1 dB (DCT). Figure 7 makes the recovery visible against the pixel-domain failures of Figure 5 (right).

Sweeping all four dataset/ratio cells shows the transform is not a free lunch, however. On MNIST at ratio 0.5 — where the pixel domain is already genuinely sparse (≈ 81% exact zeros) — the DCT *hurts* the structured learners (ALISTA SSIM 0.81 → 0.66): pen strokes are more localized in pixel space than in frequency. Recovery succeeds in whichever domain the signal is actually sparsest — the representation choice is itself part of the signal model — while LISTA-L2 remains far less domain-sensitive than the structured methods (SSIM 0.81–0.88 in the DCT domain; its worst-case DCT drop is 5.5 dB / 0.14 SSIM, on MNIST at ratio 0.5), consistent with its data-driven character.

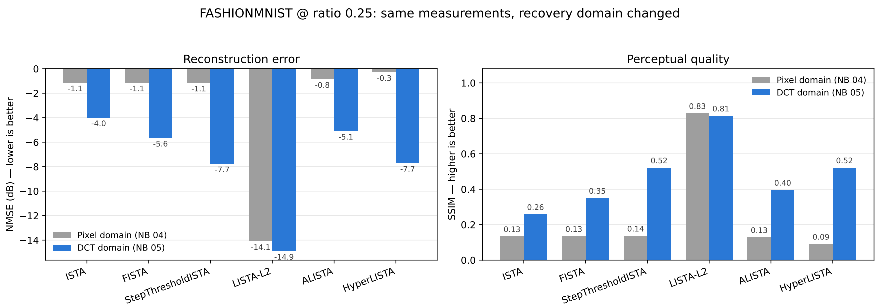

Figure 6: Same measurements, recovery domain changed (Fashion-MNIST, ratio 0.25): NMSE (left, lower is better) and SSIM (right, higher is better) per method — pixel domain (gray, Notebook 04) versus DCT domain (blue, Notebook 05). Every structured method improves sharply; LISTA-L2 is essentially domain-indifferent.

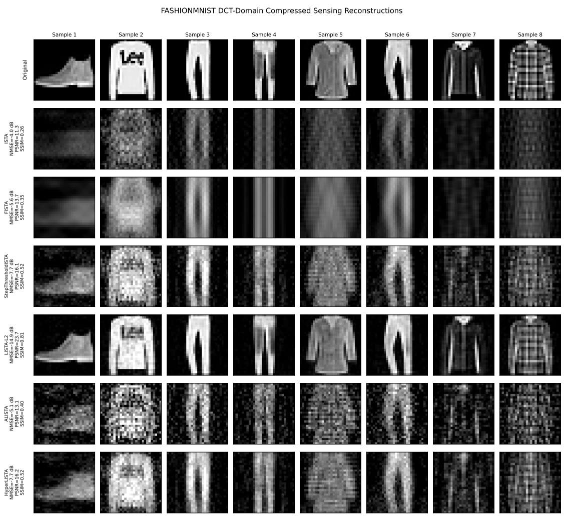

Figure 7: DCT-domain reconstructions on the failure case (Fashion-MNIST, ratio 0.25; rows as in Figure 5). Compare with the right panel of Figure 5: from the same measurements, StepThresholdISTA and HyperLISTA now recover recognizable garments (SSIM 0.52), and ALISTA no longer erases the signal.

5 Discussion

Bottom line. In practical terms, our experiments support three operating rules. (i) If the signal truly follows a known sparse model, use a HyperLISTA-style method: three tuned scalars, no gradient training, and the best accuracy by a wide margin. (ii) If the model holds but only a small depth budget is available, learning 16–32 scalars (step sizes, and secondarily thresholds) recovers LISTA-level accuracy at negligible parameter cost — this is the regime of our StepISTA/StepThresholdISTA designs. (iii) If the signal model is uncertain — e.g., real images — prefer a flexible learned model such as LISTA with deep supervision, because model-based shortcuts fail abruptly, not gracefully, under mismatch.

Structure wins when assumptions match. On data that satisfies the Gaussian-sparse model, adding structure simultaneously removed parameters and improved accuracy, culminating in HyperLISTA's -61.4 dB from 3 grid-searched scalars — an almost 40 dB advantage over the best 6M-parameter LISTA variant. This is the MBDL thesis in its purest form: when the model is right, learning should be confined to the few quantities the theory cannot fix.

The step-size schedule is what matters. Our ablation localizes the value of learning in unrolled ISTA: 16 learned step sizes recover essentially all of LISTA's gain, while 16 learned thresholds recover almost none. This offers practical guidance for designing unrolled networks under tight parameter budgets — and explains *why* ALISTA-style architectures work: they keep exactly the components (per-layer γ_k, θ_k) that matter.

Wrong prior is worse than no prior. Part B is deliberately adversarial to the structured methods, and they fail as theory predicts. This is an empirical instance of a point emphasized in the course text [7]: the statistical models underlying model-based methods are *surrogate approximations* of the true data distribution, and such surrogates "can be quite unfaithful to the true data" — e.g., a Gaussian prior permits negative values that image pixels never take. The practical lesson is not that HyperLISTA is fragile, but that its guarantees are conditional on the signal model; deploying a model-based method requires validating the model. A mismatch does not degrade performance gracefully by default — an adaptive threshold calibrated to the wrong residual statistics actively deletes signal. Section 4.3 closes the loop: the *same* measurements re-expressed over $D = A\Psi^T$ restore every structured method on Fashion-MNIST, and the reverse effect on MNIST at ratio 0.5 shows the sensitivity cuts both ways — validating a model-based method means validating its representation domain.

Limitations. (i) Both parts use a fixed sensing matrix for training and testing; generalization across matrices was not evaluated. (ii) All experiments are noiseless; a systematic noise study remains future work. (iii) Our HyperLISTA uses batch-median aggregation for $\theta^{(k)}, p^{(k)}$ rather than fully per-sample rules, and its c_3 search range had to be widened for images (optimal $c_3 \approx 0.17$ in all four pixel-domain cells, far below the synthetic-case range) — itself evidence of the model mismatch, resolved by the DCT domain where c_3 returns to 2.5–15.8. (iv) Single-seed results; error bars were not computed.

Future work. With the DCT-domain transfer realized in Section 4.3, natural continuations are: richer sparsifying transforms (wavelets, learned dictionaries) to close more of the remaining gap to LISTA-L2; image-aware thresholding for bounded, non-negative signals (DC protection, non-negativity projection); and multi-seed noise studies with error bars.

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